

Code-Layout, Readability and Reusability

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed Python indentation standards
2	Proper File Structure	Files and folders are logically organized	Yes	Organized into frontend, backend, models, database, and reports
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Used descriptive variable names
4	Function / Method Names	Functions are descriptively named	Yes	Function names clearly indicate their purpose
5	Code Comments	Inline and block comments explain logic	Yes	Important functions are documented
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for transcription, analysis, scoring, and reporting
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code optimized to avoid duplication
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles invalid audio files and processing errors

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Speech Transcription Module	Python, Whisper	Audio Processing	High

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2	Semantic Analysis Module	Python, Sentence-BERT	Concept Evaluation	High
3	Audio Feature Extraction Module	Python, Librosa	Speech Analysis	High
4	PDF Report Generator	Python, ReportLab	Report Generation	Medium
5	Streamlit User Interface	Streamlit	User Interaction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure
Readability	5	Clear naming conventions and comments
Reusability	5	Modular components reused across the application
Documentation / Comments	4	Important modules are documented
Overall Score	4.8/5	Clean, modular, and maintainable code